

PHASE-I MINI PROJECT REPORT

Project Title

E-Commerce Sales Performance Analysis Using Exploratory Data Analysis (EDA)

Objective

The primary objective of this project is to analyze e-commerce transaction data and extract meaningful business insights using Exploratory Data Analysis (EDA) techniques. The study focuses on understanding sales performance, customer purchasing behavior, product category trends, seller contributions, and payment preferences. The insights generated from the analysis can support data-driven decision-making and help improve overall business performance.

Dataset Description

This project utilizes the Brazilian E-Commerce Public Dataset (Olist) available on Kaggle. The dataset contains comprehensive information related to customers, orders, products, sellers, payments, reviews, and product categories.

Since the information is distributed across multiple tables, relevant datasets were integrated using common identifiers to create a unified analytical dataset. This consolidated dataset enabled a detailed examination of various business dimensions and performance indicators.

Data Preprocessing and Cleaning

Before conducting the analysis, extensive data cleaning and preprocessing were performed to ensure data quality and reliability.

The following activities were carried out:

- Identification and treatment of missing values.
- Detection and removal of duplicate records where applicable.
- Conversion of date-related attributes into appropriate datetime formats.
- Merging of multiple datasets using common keys.
- Validation of data consistency and integrity through quality checks.

These preprocessing steps ensured that the dataset was accurate, structured, and suitable for further analysis.

Analysis and Key Findings

The exploratory analysis provided several valuable insights into the e-commerce business operations.

Product Category Performance

A limited number of product categories were found to contribute a significant share of the overall revenue. This indicates that certain categories drive business growth and deserve strategic attention.

Geographic Sales Distribution

Sales were highly concentrated in major urban regions. Large cities generated the highest revenue, highlighting the importance of metropolitan markets in the overall sales ecosystem.

Customer Purchasing Behavior

The analysis revealed that most customers made only a single purchase. However, a relatively small group of repeat customers contributed substantially to total revenue, emphasizing the importance of customer retention strategies.

Payment Preferences

Among all payment methods, credit card transactions were the most frequently used. This suggests strong customer preference for digital payment options and highlights the need for seamless online payment experiences.

Seller Performance

A small number of sellers accounted for a significant proportion of total sales. Identifying and supporting these high-performing sellers can help strengthen the platform's overall business performance.

Customer Review Analysis

Review score analysis showed that some product categories consistently received high customer satisfaction ratings, while others displayed lower ratings and may require quality improvements or better customer support.

Visualizations Used

The following visualizations were created to support the analysis:

1. Revenue by Product Category (Bar Chart)
2. Sales by City (Bar Chart)
3. Payment Method Distribution (Bar Chart)
4. Top Seller Performance Analysis (Bar Chart)
5. Monthly Sales Trend (Line Chart)
6. Customer Review Score Comparison (Bar Chart)

These visualizations helped identify trends, patterns, and business opportunities more effectively.

Conclusion

The project successfully demonstrates how Exploratory Data Analysis can transform raw e-commerce data into actionable business intelligence. The findings reveal opportunities to increase revenue by focusing on high-performing product categories, expanding marketing efforts in top-performing cities, and implementing strategies to improve customer retention. Furthermore, improving products and services associated with lower review ratings can enhance customer satisfaction, while strengthening relationships with top-performing sellers can contribute to sustained business growth. Overall, this project illustrates the complete analytics workflow, beginning with data preparation and ending with practical business recommendations that can support strategic decision-making.